Ajay Tiwari

Senior AI/ML & Generative AI Engineer

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PROFILE SUMMARY

Senior Generative AI Engineer with 7 years of experience designing and deploying production-grade AI solutions, specializing in LLM-powered applications, agentic AI, and Retrieval-Augmented Generation (RAG). Experienced in developing a Marketing AI platform featuring conversational AI and Pulse Insights to generate actionable insights from database-driven data. Built agentic workflows integrating RAG, SQL agents, upstream systems, automated chart generation, and downloadable reports to enable natural-language data exploration and intelligent business insights. Skilled in AWS Bedrock, LangGraph, AutoGen, prompt engineering, and scalable AI application development using FastAPI, Docker, and Kubernetes.

Anthropic-certified Claude Certified Architect – Professional, with validated expertise in designing and leading enterprise-scale Claude deployments, architecting integrations, and applying governance and responsible-deployment practices.

TECHNICAL SKILLS

Programming Languages: Python, SQL, C

Generative AI: RAG, SQL RAG (Text-to-SQL), Agentic AI, Model Context Protocol (MCP), Agent2Agent (A2A) Protocol, Amazon Bedrock, Gemini, LangChain, LangGraph, LangSmith, AutoGen, Vector Databases, LLM Fine-tuning (LoRA, QLoRA)

Machine Learning and Deep Learning: Neural Networks (ANN, CNN, RNN), Transformers, Decision Trees, Random Forest, Logistic Regression, SVM, XGBoost

Frameworks: TensorFlow, PyTorch, OpenCV, Scikit-learn, Pandas, NumPy, FastAPI, Seaborn, Matplotlib

Cloud Platforms: AWS (CloudWatch, Bedrock)

Deployment: Docker, Kubernetes

Certifications: Claude Certified Architect – Professional (Anthropic, 2026)

WORK EXPERIENCE

Accenture February 2026 – Present

Lead AI/ML & Generative AI Engineer

- Designed and built the conversational layer of a GenAI chat application on AWS, using Amazon Bedrock foundation models to deliver context-aware, natural-language answers.
- Implemented RAG pipelines in the chat flow to retrieve relevant context and ground LLM responses, improving answer relevance and reducing hallucinations.
- Built a SQL RAG (text-to-SQL) capability that converts natural-language questions into SQL queries over structured data and returns the results as conversational answers.
- Integrated upstream services to collect and consolidate the data feeding the retrieval and SQL layers, keeping chat responses accurate and up to date.
- Applied Model Context Protocol (MCP) to connect agents with external tools and data sources, and the Agent2Agent (A2A) protocol to enable multiple agents to collaborate within agentic workflows.

Audax Labs

Senior Consultant (AI/ML) | January 2025 – January 2026 (Rejoined)

- Designed and implemented an OCR and data extraction system for financial documents, combining a Vision-Language Model (Qwen2.5-VL) with classical ML to extract structured data from PDFs, images and scans.
- Developed a smart bus safety system on NVIDIA Jetson Nano that detects standing students while the bus is moving, triggers alerts, and tracks boarding and deboarding counts in real time.

Senior Consultant (AI/ML) | December 2021 – August 2024

- Designed and implemented a RAG-based chatbot for an electric bike manufacturer, achieving a 20% reduction in customer care costs.
- Developed and deployed a patient-assessment chatbot using Azure OpenAI and Python, reducing patient intake time by 30%.

WalkingTree Technology August 2024 – December 2024

Senior Software Engineer (AI/ML)

- Designed and built a stock analysis agent using LangGraph that generates data-driven stock suggestions.
- Automated strategy-based code generation and execution, with self-correcting error resolution using AutoGen.

Asadel Technologies December 2020 – December 2021

Machine Learning Engineer

- Built a deep learning image-classification model that reached 86% accuracy and reduced the error rate by 12%.
- Identified traffic patterns and optimized model parameters for speed and accuracy.

Bharat Aero July 2019 – December 2020

Software Developer AI/ML

- Developed a deep learning-based object detection model to identify and track multiple objects in real-time video streams.

PROJECTS

Marketing AI for Agentic LLM Engineering

Technology Stack: LangChain, LangGraph, openai-agents, AWS Bedrock, AWS, Python, RAG, SQL RAG, Eval, Raindrop, Temporal, Langfuse, CloudWatch

- Conversational Experience: Built the core chat flow on Amazon Bedrock, supporting both document-based RAG and SQL RAG within a single chat experience.
- RAG and SQL RAG: Implemented retrieval-augmented generation for unstructured knowledge and text-to-SQL retrieval for natural-language queries on structured data.
- Upstream Data Integration: Collected and consolidated data from upstream services to power retrieval and keep responses accurate.

RAG Assistant – Generative AI for Multimedia Retrieval and Q&A

Technology Stack: Llama 3.1, vLLM, LangChain, Ragas, FastAPI, Docker

- Multimedia Processing and Embeddings: Developed a pipeline using Whisper for audio/video transcription and OCR for text extraction from images, converting extracted text into vector representations (embeddings) and storing them in ChromaDB for faster retrieval.
- Intelligent Retrieval and Response Generation: Built a retrieval-generation chain where the Retriever fetches relevant documents based on user queries, and Llama3.1-8B generates comprehensive answers using the retrieved information. Evaluation with Ragas yielded 98% faithfulness and 92% answer correctness.
- Scalable Backend and Deployment: Developed a performant FastAPI backend and deployed the system using Docker containers on Kubernetes, ensuring scalability and efficient operation.

Entity Extraction – OCR and Data Extraction from Financial Documents

Technology Stack: Qwen2.5-VL, docTR, FastAPI, Docker, Table Transformer, DBSCAN

- Designed and implemented an OCR-powered financial data extraction pipeline leveraging the Qwen2.5-VL vision-language model for multi-format document understanding (PDFs, images, and scanned documents).
- Combined vision-language model outputs with classical ML-based and regex-based text classification to refine the extracted entities.

CERTIFICATIONS

Claude Certified Architect – Professional | Anthropic | Issued: October 2026 | Expires: October 2027

Credential: https://www.credly.com/badges/5bf21504-3fe0-453e-b86d-23c98dc086f3

EDUCATION

B.Tech (ECE), Bundelkhand Institute of Engineering & Technology Aug 2015 – 2019